BLUESTOCK MUTUAL FUND CAPSTONE PROJECT

Internship Report |Submitted by :- vprvithal

ABSTRACT

The Bluestock Mutual Fund Analytics Capstone Project was developed to perform end-to-end financial data analysis using modern data analytics technologies. The project focused on transforming raw mutual fund datasets into meaningful business insights using Python, SQLite, SQL, and Power BI.

The project began with extracting multiple raw datasets containing mutual fund information, benchmark indices, investor transactions, portfolio holdings, industry statistics, and SIP inflow data. These datasets were cleaned, standardized, and validated through an automated ETL pipeline before being stored in a SQLite database.

After preprocessing, several financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Tracking Error, Maximum Drawdown, Historical Value at Risk (VaR), and Conditional Value at Risk (CVaR) were calculated using Python.

Advanced analytical techniques including Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Sector Concentration Analysis using Herfindahl–Hirschman Index (HHI), and a Rule-Based Fund Recommendation System were implemented to provide deeper insights into mutual fund performance and investor behavior.

Finally, an interactive Power BI dashboard consisting of four business-focused pages was developed to visualize industry trends, fund performance, investor analytics, and SIP market trends.

The project successfully demonstrates the complete data analytics lifecycle including data engineering, financial analytics, database management, business intelligence, and reporting.

TABLE OF CONTENTS

INTRODUCTION

Mutual funds have become one of the most popular investment options for retail and institutional investors due to their diversification, professional management, and accessibility. Every day, large volumes of financial data are generated, including Net Asset Values (NAVs), investor transactions, benchmark indices, portfolio holdings, and industry statistics.

Analyzing this data manually is difficult because of its size and complexity. Financial institutions require automated systems capable of cleaning, processing, analyzing, and visualizing large datasets to support informed investment decisions.

The Bluestock Mutual Fund Analytics Project was developed to address this need by building an end-to-end analytics solution. The project integrates data engineering, financial analysis, database management, and business intelligence into a single workflow.

Using Python, SQL, SQLite, and Power BI, raw mutual fund data was transformed into meaningful business insights that can help investors, analysts, and financial institutions understand market trends, evaluate fund performance, measure investment risks, and analyze investor behavior.

This project follows the complete Data Analytics Lifecycle beginning from raw data ingestion and ending with an interactive Power BI dashboard.

TECHNOLOGY STACK

Technology	Purpose
Python	Data preprocessing and analytics
Pandas	Data manipulation and cleaning
NumPy	Mathematical calculations
SQLite	Database storage
SQL	Data querying
Power BI	Dashboard development
Git & GitHub	Version control
Jupyter Notebook	Analysis and documentation

PROJECT ARCHITECTURE

```
bluestock_mf_capstone/
│
├─── data/
│    ├─── raw/
│    ├─── processed/
│    └─── db/
│         └─── bluestock_mf.db
│
├─── notebooks/
│    ├─── 01_data_ingestion.ipynb
│    ├─── 02_data_cleaning.ipynb
│    ├─── 03_eda_analysis.ipynb
│    ├─── 04_performance_analytics.ipynb
│    └─── 05_advanced_analytics.ipynb
│
├─── scripts/
│    ├─── etl_pipeline.py
│    ├─── live_nav_fetch.py
│    ├─── compute_metrics.py
│    └─── recommender.py
│
├─── sql/
│    ├─── schema.sql
│    └─── queries.sql
│
├─── dashboard/
│    └─── bluestock_mf_dashboard.pbix
│
├─── reports/
│    ├─── Final_Report.pdf
│    ├─── Presentation.pptx
│    └─── charts/
│
├─── README.md
```

ETL PIPELINE

ETL stands for **Extract, Transform, and Load**. It is the process of collecting raw data, cleaning and transforming it into a usable format, and finally storing it in a database for further analysis.

ETL WORKFLOW

RAW CSV FILES
|
▼
Read using Pandas
|
▼
Remove Duplicate Rows
|
▼
Handle Missing Values
|
▼
Standardize Column Names
|
▼
Convert Date Columns
|
▼
Save Cleaned CSV Files
|
▼
Load into SQLite Database
|
▼
Ready for Analytics

- Transform Phase
- Cleaning Operations Performed
- Duplicate Removal
- Missing Value Handling
- Standardizing Column Names
- Date Conversion
- Data Type Validation

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) is the process of understanding the characteristics of a dataset before performing advanced analytics or machine learning. It helps identify trends, patterns, missing values, correlations, and outliers.

In this project, EDA was performed using **Python**, **Pandas**, **NumPy**, and **Matplotlib**. Various visualizations were created to understand mutual fund performance, investor behavior, and industry growth.

The objective of EDA was to ensure that the cleaned data was suitable for financial analysis and dashboard development.

Statistics

Summary statistics were generated to understand the distribution of numerical variables.

The following statistical measures were calculated:

- Count
- Mean
- Median
- Standard Deviation
- Minimum Value
- Maximum Value
- Quartiles

Distribution Analysis

Time Series Analysis

Correlation Analysis

POWER BI DASHBOARD

Power BI was used to transform analytical results into an interactive business intelligence dashboard. The dashboard enables users to explore mutual fund performance, investor behaviour, and market trends using interactive charts, KPI cards, and slicers.

Dashboard Page 1 – Industry Overview

This dashboard provides an overview of the Indian Mutual Fund Industry.

Visuals Included

- KPI Cards
- Industry AUM Trend
- AUM by Fund House
- Slicers

Dashboard Page 2 – Fund Performance

This dashboard compares the financial performance of individual mutual funds.

Visuals Included

- Scatter Plot
- Fund Scorecard
- NAV Trend
- Benchmark Comparison
- Interactive Filters

Dashboard Page 3 – Investor Analytics

This dashboard analyzes investor behaviour using transaction history.

Visuals Included

- State-wise Investment
- SIP vs Lumpsum Distribution
- Age Group Analysis

- Monthly Transactions
- Slicers

Dashboard Page 4 – SIP & Market Trends

This dashboard analyzes SIP inflows and compares them with market performance.

Visuals Included

- SIP Inflow Trend
- NIFTY 50 Comparison
- Category Inflow Heatmap
- Top Categories by Net Inflow

The following business insights were obtained from the analysis:

1. The Indian Mutual Fund Industry experienced continuous growth in Assets Under Management throughout the study period.
2. Liquid funds demonstrated stable returns with comparatively lower investment risk.
3. Small-cap equity funds generated higher returns but were associated with increased volatility.
4. Mutual funds with higher Sharpe Ratios consistently delivered superior risk-adjusted performance.
5. Positive Alpha values indicated that several actively managed funds outperformed their benchmark indices.
6. Historical VaR and CVaR effectively measured downside investment risk during adverse market conditions.
7. Rolling Sharpe Ratio highlighted changing fund performance over different market periods.
8. Investor Cohort Analysis revealed increasing participation from newer investors.
9. SIP Continuity Analysis identified investors at risk of discontinuing their SIP investments.
10. Sector HHI analysis showed that diversified portfolios carried lower concentration risk than highly concentrated portfolios.

CONCLUSION

The Bluestock Mutual Fund Analytics Capstone Project successfully demonstrated the complete data analytics lifecycle from raw data collection to business intelligence reporting.

The project integrated Python, SQL, SQLite, and Power BI to automate data preprocessing, calculate financial performance metrics, analyze investment risk, understand investor behaviour, and visualize insights through interactive dashboards.

The ETL pipeline ensured high-quality processed datasets, while the analytical modules generated meaningful financial indicators such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Historical VaR, and Rolling Sharpe Ratio.

Advanced analytics including Investor Cohort Analysis, SIP Continuity Analysis, Sector HHI, and the Rule-Based Fund Recommendation System provided additional business value beyond traditional financial reporting.

The Power BI dashboard transformed analytical outputs into an easy-to-understand decision support system for investors and financial analysts.

Overall, this project strengthened practical skills in Data Engineering, Financial Analytics, Database Management, Business Intelligence, and Data Visualization while demonstrating the ability to solve real-world business problems using modern analytics tools.

THANK YOU

Bluestock Mutual Fund Capstone Project